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ME5250 – Project 1

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1. Mechanism Overview

This project investigates a 3-DOF parallel wrist mechanism designed for controlled spatial orientation. Three kinematic legs connect a fixed base to a moving outer race in a closed-loop structure. Two legs are actuated via coplanar cylindrical joints oriented orthogonally, providing coupled rotational and translational motion, while the third uses a centrally located revolute input for rotation about the vertical axis. Passive universal and spherical joints constrain the system to pure rotation about a fixed point, yielding a spherical parallel mechanism with improved stiffness and load distribution compared to a serial gimbal equivalent.

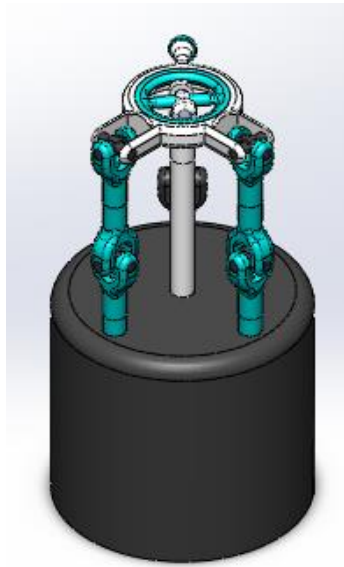


Figure 1: CAD model of mechanism

Video Analysis Link: <https://tinyurl.com/5ak94uz6>

2. Literature Review

Parallel wrist mechanisms have been widely studied due to their high stiffness, accuracy, and load-carrying capability. A foundational example is the Stewart Platform, which provides six degrees of freedom using six extensible legs [1]. Reduced-DOF variants, particularly spherical parallel wrists, have been developed for applications requiring precise orientation control.

Research has explored configurations such as 3-RRR spherical wrists and hybrid joint architectures incorporating universal and spherical joints to simplify construction while maintaining kinematic constraints [2]. These mechanisms are often analyzed using loop-closure

equations and constraint-based formulations due to the absence of a straightforward serial kinematic chain.

Some key challenges identified in prior work include nonlinear forward kinematics, multiple solution branches, the presence of singular configurations, and coupled actuator behavior.

3. Construction Overview

The mechanism explored in this project was adapted from a homework problem requiring 3 degrees of freedom [3]. It consists of a rigid base, three articulated legs, and a moving outer race, with each leg combining active and passive joints to enforce kinematic constraints while allowing controlled motion.

Bearing: The outer and inner race are connected by 1 revolute joint. Universal joints attach at a the radius, r , which is 80mm.

Cylindrical Input Legs (Legs 1 & 2): The two cylindrical input legs are oriented orthogonally to decouple their motion. Each contains 2 universal joints and 1 cylindrical joint. The upper universal joint connects the yoke to the outer race via male and female pins, while the lower universal joint links to a shaft extension that forms the cylindrical joint with the base plate.

Rotational Input Leg (Leg 3): The rotational input leg connects to the spherical center via a universal yoke attached to the pin bisecting the inner race. It contains 1 revolute joint — between the base plate and the shaft extension — and 1 universal joint.

Socket: The spherical socket, positioned away from each cylindrical leg, contains 1 spherical joint shared with the outer race, 1 prismatic joint, and 1 revolute joint between the socket and base plate.

4. Degree of Freedom Analysis (DOF)

The DOF of the mechanism can be analyzed using the Grübler equation for spatial mechanisms:

$$DOF = m(N - 1 - J) + \sum_{i=1}^J f_i$$

where m is the total DOF of the space, N is the number of links (including 1 common ground), J is the total number of joints, and f_i is the DOF of each joint. The Grübler DOF analysis of the parallel wrist mechanism is shown below:

$$6(10 - 1 - 12) + 21 = 3 \text{ DOF}$$

The 3 DOF correspond to the 3 independent rotational DOF of the mechanism. The table below tracks the contributions from each joint type:

Table 1: Joint count and DOF contributions

		f_i	Leg 1	Leg 2	Leg 3	Socket	Bearing	# of Joints	$\sum f_i$
Joint Type	Revolute (R)	1			1	1	1	3	3
	Prismatic (P)	1				1		1	1
	Cylindrical (C)	2	1	1				2	4
	Universal (U)	2	2	2	1			5	10
	Spherical (S)	3				1		1	3
Totals								12	21

5. Range of Model Inputs and Self-Intersection

The mechanism is driven by:

$$q = [\theta_1, \theta_2, \theta_3]$$

where θ_1 is the translation of the first cylindrical input along the y-axis, θ_2 is the translation of the second cylindrical input along the y-axis, and θ_3 is the rotational input about the y-axis. The home position of the parallel wrist mechanism is defined as the position of the joints that allow the end effector to be parallel to the base plate.

With actuator limits:

$$|\theta_1| \text{ \& \; } |\theta_2| \leq 57\text{mm from the home position}$$

These limits were determined by analyzing the motion of the mechanism in SOLIDWORKS Motion Analysis using the contact and interference tools. Ignoring these limits, the following graph of the theoretical workspace was generated using MATLAB:

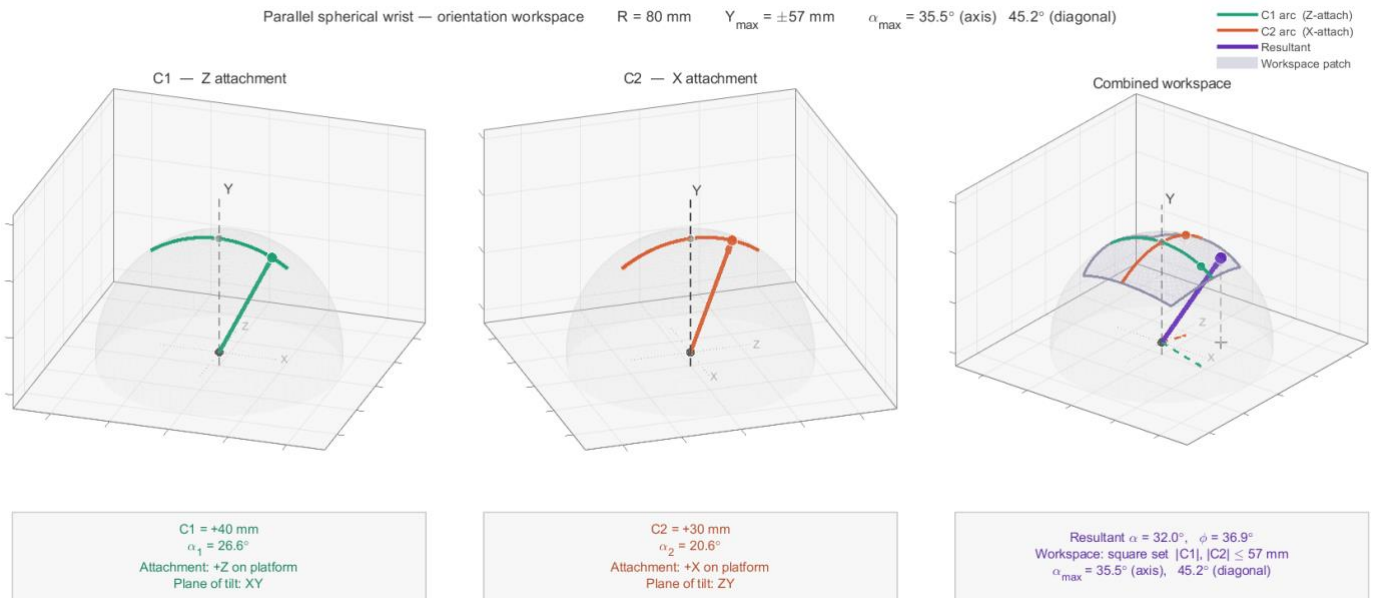


Figure 5: Workspace boundaries

6. Degree Coupling

The mechanism exhibits strong coupling between actuator inputs and output orientation. Each actuator influences multiple rotational components due to closed-loop constraints. θ_1 primarily affects Z-axis rotation, θ_2 primarily affects X-axis rotation, and θ_3 controls Y-axis rotation. However, motion is nonlinear and interdependent.

7. Forward Kinematics

The forward kinematics describe the mapping from actuator inputs to platform orientation.

Although the mechanism contains multiple joint types (R, C, U, S), the forward kinematics can be expressed solely in terms of the platform orientation due to the spherical constraint imposed by the passive joints. These joints do not introduce additional degrees of freedom but instead enforce geometric constraints, allowing the system to be modeled as a 3-DOF rotational mechanism. From geometry the end effector angles θ_z and θ_x are determined below:

$$\theta_z = \arcsin\left(\frac{\theta_1}{r}\right), \theta_x = \arcsin\left(\frac{\theta_2}{r}\right), \& \theta_y = \theta_3$$

The rotation matrix is:

$$R = R_y(\theta_y)R_z(\theta_z)R_x(\theta_x) = \begin{bmatrix} c\theta_y & 0 & s\theta_y \\ 0 & 1 & 0 \\ -s\theta_y & 0 & c\theta_y \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & c\theta_z & -s\theta_z \\ 0 & s\theta_z & c\theta_z \end{bmatrix} \begin{bmatrix} c\theta_x & -s\theta_x & 0 \\ s\theta_x & c\theta_x & 0 \\ 0 & 0 & 1 \end{bmatrix}$$
$$R = \begin{bmatrix} c\theta_z c\theta_y & -s\theta_z & c\theta_z s\theta_y \\ s\theta_x s\theta_y + c\theta_x s\theta_z c\theta_y & c\theta_x c\theta_z & c\theta_x s\theta_z s\theta_y - s\theta_x c\theta_y \\ s\theta_x s\theta_z c\theta_y - c\theta_x s\theta_y & s\theta_x c\theta_z & c\theta_x c\theta_y - s\theta_x s\theta_z s\theta_y \end{bmatrix}$$

where s and c , represent $\sin()$ and $\cos()$ respectively.

8. Singularity Analysis

Singularities occur when $\det(J) = 0$. In this mechanism a singularity would occur if $|\theta_1|$ or $|\theta_2|$ were to exceed the radius, $r = 80\text{mm}$. Since the actuator limits of $\pm 57\text{mm}$ are less than $r = 80\text{mm}$, the $\arcsin()$ argument never reaches ± 1 and no singularity is reachable within the workspace. However, due to the geometric constraints of the design and interference points, the mechanism is unable to reach any points where a singularity would occur.

9. References

1. https://scienceinsights.org/what-is-a-stewart-platform-and-what-is-it-used-for/#google_vignette
2. <https://robotics.caltech.edu/~jwb/courses/ME115/handouts/AgileEyeGosselin.pdf>
3. ME5250 – Homework Set 2, Problem 2 (Parallel Wrist Mechanism)